Lam and Horváth Reply: In a previous experiment [1], imbibition fronts moving up in a paper sheet, which drifts down into a water tank at speed v, were found to have an average height $\bar{h} \sim v^{-1/\Omega}$, a width $w \sim \bar{h}^{\Omega\kappa}$, and a temporal correlation function $C(t) = v^{-\kappa} f(t/v^{(\theta_t+\kappa)/\beta})$. We simulated the phenomenon using a pipe network model and reproduced the rich scaling behaviors with *all* exponents Ω, κ, θ_t, and β in agreement with the experimental values [2]. Another study based on a phase-field model leads to predictions on exponents that are unfortunately incompatible with the experimental ones [3].

The front propagation exemplifies two related dynamics, namely, mean flow and roughening. Assuming a unique time scale controlling both dynamics, an exponent identity $\beta = \Omega(\theta_t + \kappa)/(\Omega + 1)$ was derived. Inserting experimental values into the right-hand side, we get $\beta = 0.52$ compared with the measured value 0.56. Using numerical estimates, $\beta = 0.54$ is obtained to be compared to the numerically measured value 0.63. The identity is also supported by a constancy in the relative degree of roughening for simulated fronts in stationary paper sheets [2].

The unique time scale assumption and hence the identity are the main objections by Dubé *et al.* [4]. We now give further evidences by directly extracting relevant time constants from experimental and numerical data in Refs. [1,2]. The mean flow is characterized by $t_1 = \bar{h}/v$. For roughening, we observe that $C(t)$ can be nicely fitted by $\sqrt{2}\, w[1 + (t/t_2)^{-a\beta}]^{-1/a}$ at $a = 3.5$ and the relaxation time t_2 hence follows. Results are plotted in Fig. 1. The unique time scale assumption means that t_1 and t_2 are different representations of a unique time scale for the front propagation dynamics. It requires t_1/t_2 to be a constant of order, but *not* necessarily exactly unity. In contrast, if flow and roughening were distinct dynamics [3], t_1 and t_2 could easily differ by a factor of, say, 10 or 100. Using the experimental results, $t_1/t_2 \approx 0.82$ on average for various values of $\bar{h}$. It is close to unity and is a strong experimental support of our view. A further numerical support is that $t_1/t_2 \approx 2.0$ roughly from our simulation. The very similar magnitudes of t_1 and t_2 *both* in experiment and simulation independently, in addition to their approximate proportionality, should not be casually dismissed as accidental. Instead, its implication of a unique time scale can be a helpful hint for a complete theory of the experimental observations.

The discrepancy between the ratios 0.82 and 2.0 deserves some caution, though it can be due to finite size effects, which affects prefactors more severely than exponents. In Fig. 1, variations in the slopes of the fitted lines for either the experimental or numerical values reflect slight deviations from $t_1 \propto t_2$. These are alternative representations of deviations of the estimated exponents from our identity discussed above. They have been attributed to finite size effects which are particularly significant in the simulations [2].

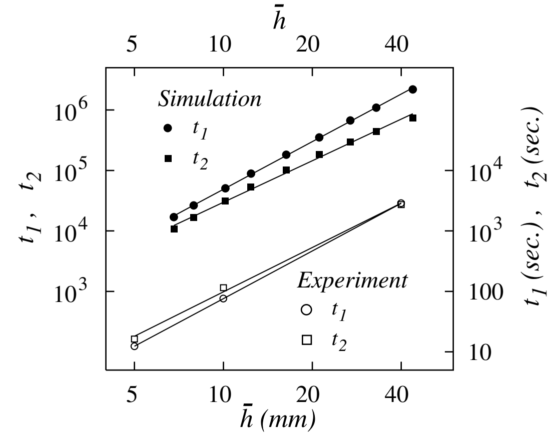

FIG. 1. Log-log plot of characteristic time t_1 and t_2, respectively, for flow and roughening against interface height $\bar{h}$ from experiment (open symbols, right and bottom axes) and simulation (solid symbols, left and top axes).

C. H. L. is supported by HK RGC Grant No. 5191/99P. V. K. H. is supported by Hungarian Science Foundation Grants No. OTKA-F17310, No. M-27950, and NATO Grant No. DGE-9804461.

Chi-Hang Lam[1] and Viktor K. Horváth[2]
[1]Department of Applied Physics
Hong Kong Polytechnic University
Hung Hom, Hong Kong, China
[2]Department of Physics
University of Pittsburgh
Pittsburgh, Pennsylvania 15260
and Department of Biological Physics
Eötvös University
Pázmány P. 1A. 1117
Budapest, Hungary

Received 2 February 2001
DOI: 10.1103/PhysRevLett.86.6047
PACS numbers: 47.55.Mh, 05.40.–a, 05.70.Ln, 68.35.Fx

[1] V. K. Horváth and H. E. Stanley, Phys. Rev. E **52**, 5166 (1995).
[2] C.-H. Lam and V. K. Horváth, Phys. Rev. Lett. **85**, 1238 (2000).
[3] M. Dubé *et al.,* Phys. Rev. Lett. **83**, 1628 (1999); Eur. Phys. J. B **15**, 701 (2000).
[4] M. Dubé *et al.,* preceding Comment, Phys. Rev. Lett. **86**, 6046 (2001).

